

DIPX 2023

DIPLOMA PROJECT EXHIBITION

CENTRE FOR DIPLOMA STUDIES



DCN QUIZZES APPLICATION

ABSTRACT

Data Communication and Networking is a subject in university IT courses. UTHM students struggle to access past quiz questions on the website, which lack answers. Existing quiz apps are not user-friendly or secure. The goal is to design a safe and user-friendly quiz app for UTHM students studying Data Communication and Networking. It will provide quizzes on specific topics and improve the user experience. The target users are UTHM Year 2 Semester 2 students in the IT Diploma course. Data communication modules and networks are used for quiz questions, and the ADDIE model guides the development process. Visual Studio Code is used for system development, with Firebase supporting the back-end database. A questionnaire was given to 30 UTHM students to collect data, and the overall feedback on the DCN quiz app was positive. We will continue to enhance the app for increased student usage.

OBJECTIVE

Objective 1

To design a quiz application that enables UTHM students to answer quizzes with a secure and user-friendly interface.

1

Objective 2

To develop an easy and convenient mobile quiz application by providing quizzes for UTHM students on various topics specifically only for subject Data Communication and Networking.

2

Objective 3

To evaluate the effectiveness and the usability and enhance the user experience of the application for UTHM's students.

3

METHODOLOGY

We use ADDIE Model for DCN Quizzes Application.



RESULTS & DISCUSSION

Our innovation, the quiz application, provides a convenient assessment method for students to test their knowledge anytime and anywhere. It offers real-time feedback, allowing students to understand their strengths and areas for improvement. Additionally, the application enables students to track their progress over time. These features contribute to our project objective by allowing students to evaluate their comprehension, identify weaknesses, and focus on improving specific topics. The quiz app also serves as a valuable tool for review and revision, helping students solidify their understanding. Furthermore, elements like leaderboards and score systems foster healthy rivalry among students, encouraging active participation and motivating them to achieve better results, ultimately enhancing engagement and improving learning outcomes.



USEFULNESS

We are trying to solve the problem of limited assessment options. Exams and assignments, for example, may not provide real-time feedback to effectively test students' understanding of subject content. In addition, we would like to complete the tracking and analytics of student progress. It may be difficult for instructors to assess each student's progress in a large class. Students and educators alike can assess performance, identify areas for growth and track overall learning trajectories by including a progress tracking element in our quiz app. And lastly our target is UTHM students who are in year 2 semester 2 who are taking Data Communication and Network subjects in the Information Technology course.



IMPACT / CONTRIBUTION TO SOCIETY

Students will get an A grade in the subject of Data Communication and Networking after taking the quiz found in our application. Next, students can increase their knowledge by taking this quiz. In addition, it can make it easier for lecturers who teach the subject of data communication and networking, that is, the lecturer only asks students to download the dcn quizzes application to make additional quizzes.



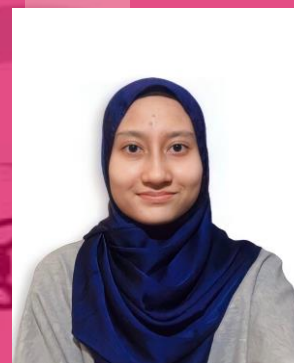
VALUE ADDED / COMMERCIALIZATION

This innovation stands out because of cooperation and competition. The advantage of our app over others is that our quiz app can be easily accessed by students anytime, anywhere, as long as they have a compatible device and an internet connection. Furthermore, interactive features such as instant feedback, progress tracking and leaderboard positions can encourage students to actively participate and strive for better performance.

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